INOVATALK 2026 | STEM | Lentera Harmoni: Green & Sustainable Innovation

IRIS - Intelligent Rice Integrated System

Rain-aware AWD, canopy-anomaly triage, and a grounded assistant - on one plot

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01 PROBLEM

In 2024 Indonesia harvested 10.05 million hectares of paddy and produced 53.14 million tonnes of milled dry unhusked grain (GKG) [1]. The usual way to grow that crop - keeping the field flooded - is also why irrigated rice is a major agricultural source of methane. IPCC AR6 assigns non-fossil CH4 a GWP100 of 27 [2].

The agronomic reply is already on the shelf. IRRI's safe AWD protocol lets the water table fall to 15 cm below the soil surface, then refills the field to about 5 cm, and holds a flood through flowering [3]. At that threshold a meta-analysis reports 23.4% less irrigation water with no meaningful yield loss [5]; adoption studies in Asia report cuts of up to 38% when the protocol is followed [4]. A greenhouse-gas meta-analysis finds CH4 down 51.6% versus continuous flooding, with N2O up 44.0% [6]. The science is not the missing piece.

What stalls in the field is the watch. Safe AWD still depends on reading a perforated field water tube [3], [4], and leaf stress is often noticed only after damage is advanced. Both jobs sit on the same plot. That is the gap software can close.

02 APPROACH

IRIS is a web application for that plot. It does not invent a new cultivation method. It runs the existing safe-AWD protocol, reads the canopy for anomalies, and answers the farmer from the same records.

1. Irrigation is scheduled by growth stage. Irrigation is deferred when the forecast is >=15 mm of rain within 72 h [12], unless the water table is already near the dry limit. A sensor ingest API is in place; field hardware has not been tested.

2. Four leaf-disease classes are recognised with MobileNetV3-Large [9] and then scored against the plot's water and weather through explicit rules. That combination is the anomaly signal - not the photo class alone. The worked example is hypothetical. No end-to-end fusion model is used.

3. The assistant answers in Indonesian from plot data and cited sources. If the language-model endpoint is unavailable, it searches the plot status text. Pesticide doses are not recommended.

03 METHODS

Water use and CH4 are computed from a 100-day water-balance simulation (0 mm rain, 0.8 cm day-1 drawdown, 1 ha). Sensor calibration and a mesocosm trial remain protocol only; no field data have been collected.

The image classifier was trained on 5,932 public photographs of four rice-leaf diseases [10], [11]. Performance on Indonesian leaves has not been measured, so test accuracy is not reported.

Emissions follow IPCC 2006 Tier 1 [7]: CH4 = EF_c · t · A. EF_base = 1.30 kg ha-1 day-1 (Table 5.11); SF_w = 1.00 (continuous flooding) or 0.78 (multiple aeration; Table 5.12). Effective SF_w 0.8922 is a linear interpolation by flooded-day fraction, a project assumption rather than an IPCC equation. GWP100 = 27 [2]. N2O is omitted (CO2e is an upper bound). The 2019 Refinement sets SF_w for AWD at 0.55 [8]; the 2006 chain used here is the more conservative claim.

04 WORKFLOW

Two observations are taken on the same plot and meet in one loop. The water process records the water table, then runs the stage scheduler and the rain-hold rule. The canopy process checks a photograph and classifies it on CPU. Both feed a rule matrix, the assistant, and an emission sheet. Pumps are not actuated; the user decides. During flowering the rain-hold rule is disabled so the flood is kept.

05 RESULTS

-37.5%

irrigation water 8,000 -> 5,000 m3/ha

CH4 -10.8% | 0.378 t CO2e/ha | 100 days, 1 ha

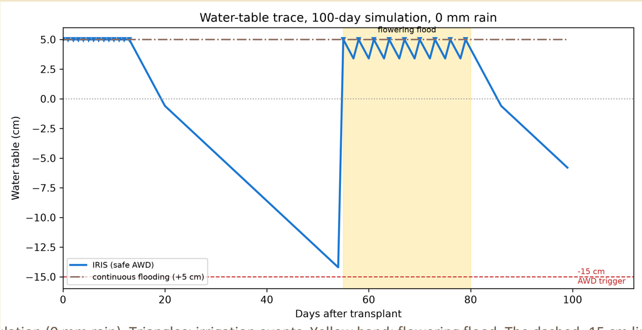

Fig. 1. Water-table trace, 100-day simulation (0 mm rain). Triangles: irrigation events. Yellow band: flowering flood. The dashed -15 cm line is the safe-AWD trigger [3]; it is rarely reached in this run.

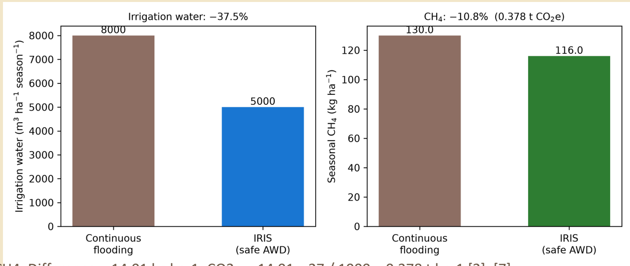

Fig. 2. Seasonal irrigation volume and CH4. Difference = 14.01 kg ha-1; CO2e = 14.01 x 27 / 1000 = 0.378 t ha-1 [2], [7].

06 NUMERICAL SUMMARY

1 ha / 100 days (water-balance simulation)

Metric	CF	IRIS
Water (m3)	8,000	5,000
Flooded days	100	51
CH4 (kg)	130.00	115.99
CO2e avoided	-	0.378 t
Model events	100*	23*

*Irrigation events are daily water-balance outputs (0.8 cm day-1), not 100 farmer irrigations and not 23 dry-down cycles to -15 cm.

07 PROTOTYPE

The web application (Next.js and FastAPI) is organised around one demonstration plot. Irrigation, canopy triage, and the assistant read the same plot record. The emission sheet shows the simulation figures in the table, not the 30-day window of the demonstration plot. The four disease classes run on CPU. Indonesian leaf photographs have not been evaluated.

08 IMPLICATIONS

On one hectare the simulation saves 3,000 m3 of water per season (-37.5%) and cuts CH4 by 10.8%. Those figures are smaller than full AWD ranges in the literature [4]-[6] because the water table rarely reached -15 cm in this run. What is available now is working software and a repeatable calculation; field measurements have not been made.

09 CONCLUSION

IRIS does not replace IRRI's safe-AWD protocol [3]. It puts that protocol, canopy-anomaly triage, and a grounded assistant on the same plot, and writes water and CH4 under IPCC Tier 1 [7]. The numbers on this poster come from a water-balance simulation. Field sensors, chamber CH4 measurements, and Indonesian leaf-image tests have not been carried out.

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